METRIC

FF-B-171/19

30 NOVEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, BALL, ANNULAR, SINGLE ROW, ANGULAR CONTACT, COUNTERBORED OUTER RING, CONTACT ANGLE ABOVE 10° UP TO AND INCLUDING 22°, NON-SEPARABLE, ANSI/AFBMA DIMENSION SERIES 19.

The complete requirements for acquiring the bearings described herein shall consist of this document and the latest issue of FF-B-171/GEN.

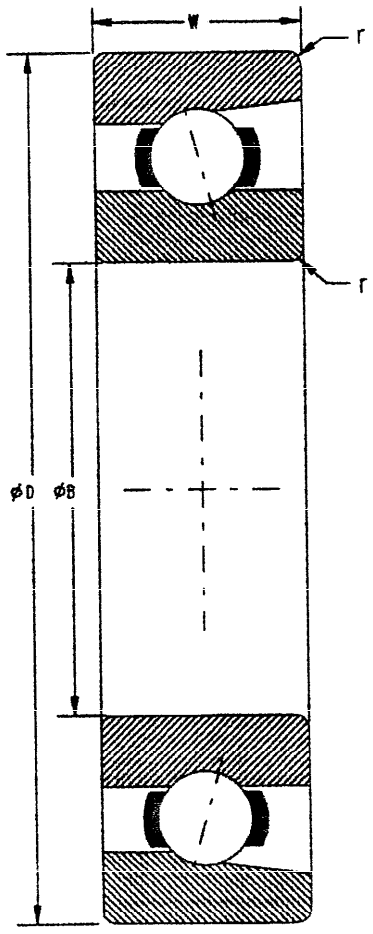


FIGURE 1.
OPEN (NO SHIELDS OR SEALS)
(CODE A)

BEARING CONFIGURATIONS

AMSC/NA

FSC 3110

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REQUIREMENTS:

1. Boundary dimensions. The boundary dimensions shall be specified in the part number by the size code in table I.

2. Cages. Cages shall be manufactured from any material unless otherwise specified. Cages shall be specified in the part number by the codes in table II.

3. Shields and Seals. All angular contact bearings with counterbored outer rings shall be open (no shields or seals, see figure 1) and designated by the code "A" in the part number.

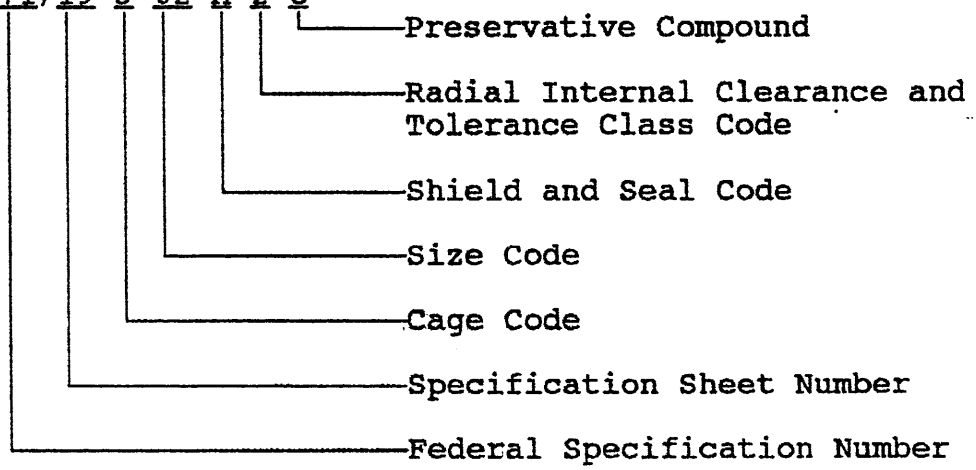
4. Radial Internal Clearance. The tolerance limits for radial internal clearance shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table III.

5. Tolerance Class. The tolerance limits for tolerance class ABEC-1 and ABEC-5 shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table III.

6. Preservative Compound. All bearings shall be supplied with preservative compound in accordance with MIL-P-197 and designated by the code "C" in the part number.

7. Part number. The part number consists of the letters FFB, the basic Federal specification number, the specification sheet number, and alpha or numeric codes for cage, size, shields and seals, radial internal clearance and tolerance class, and preservative compound.

EXAMPLE: FFB171/19 J 02 A 2 C



FFB171/19J02A2C indicates - Bearing, cage - pressed steel or molded plastic; bore 15mm; outside diameter 28mm; width 7mm; open (no shields or seals); radial internal clearance symbol 0 (normal); tolerance class ABEC-1; preservative compound in accordance with MIL-P-197.

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TABLE I. Boundary dimensions.

Size Code 1/	OB Bore	OD Outside Diameter	W Width	Chamfer $r_s$min 2/	Shaft Shoulder Diameter Min 3/	C Dynamic Load Rating (Pounds)
00	10	22	6	0.3	12	480
01	12	24	6	0.3	14	510
02	15	28	7	0.3	17	760
03	17	30	7	0.3	19	800
04	20	37	9	0.3	22	1480
05	25	42	9	0.3	27	1190
06	30	47	9	0.3	32	1480
07	35	55	10	0.6	39	1730
08	40	62	12	0.6	44	2205
09	45	68	12	0.6	49	2320
10	50	72	12	0.6	54	2985
11	55	80	13	1.0	60	3495
12	60	85	13	1.0	65	3670
13	65	90	13	1.0	70	3730
14	70	100	16	1.0	75	5120
15	75	105	16	1.0	80	5210
16	80	110	16	1.0	85	5300
17	85	120	18	1.1	91.5	6915
18	90	125	18	1.1	96.5	8095
19	95	130	18	1.1	101.5	7165
20	100	140	20	1.1	106.5	9000
21	105	145	20	1.1	111.5	9170
22	110	150	20	1.1	116.5	8400
24	120	165	22	1.1	126.5	11300
26	130	180	24	1.5	138	13725
28	140	190	24	1.5	148	13895
30	150	210	28	2.0	159	19190
32	160	220	28	2.0	169	19470
34	170	230	28	2.0	179	20225
36	180	250	33	2.0	189	28565
38	190	260	33	2.0	199	26260
40	200	280	38	2.1	211	33060

1/ Size codes in Table I are industry bore codes.

2/ The minimum bearing chamfer ($r_s$min) must clear the maximum shaft or housing fillet radius. This dimension does not control the shape of the bearing chamfer contour.

3/ For reference purposes only. Maximum shoulder heights shall be such as to permit bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, provisions shall be made for the removal of the bearing from the shaft without damage to the bearing or other parts.

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TABLE II. Codes for Cages.

Code	Cage
J	Pressed steel or molded plastic
M	Machined bronze or brass
T	Machined non-metallic (phenolic)
Y	Pressed brass

TABLE III. Codes for Radial Internal Clearance and ABEC tolerance.

Code	Radial Internal Clearance	Tolerance Class
1	Symbol 2 (Less than normal)	ABEC-1
2	Symbol 0 (Normal)	
3	Symbol 3 (Greater than normal)	
4	Symbol 4 (Greater than symbol 3)	
5	Symbol 2 (Less than normal)	ABEC-5
6	Symbol 0 (Normal)	
7	Symbol 3 (Greater than normal)	
8	Symbol 4 (Greater than symbol 3)	

MILITARY INTERESTS:

CIVIL AGENCY
GSA-FSS

Military Coordinating Activity:
DLA-IS

PREPARING ACTIVITY:
DLA - IS

Custodians:
Army - AT
Air Force - 11

(Project 3110-0848-19

Review Activities:
Army - AR, EA, MI
Navy - OS

User Activity:
Army - GL, ME

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